

Two frontier questions in modified gravity—whether the quasi-static approximation (QSA) is exact in cuscuton theory and whether Hubble-dependent kinetic structures enable ghost-free phantom expansion—have attracted growing attention, especially after the DESI BAO results suggesting dark energy evolution. **The QSA is indeed exact in cuscuton gravity**, a conclusion supported by three independent lines of argument across multiple papers, though no single publication states this in those precise words. **No paper proposes $K \propto 1/H^2$ to achieve phantom expansion without ghosts**; the most relevant 2026 result proves a no-go theorem for single-field Hubble-modulated kinetic structures. However, the cuscuton framework itself can produce arbitrary equations of state—including $w < -1$ —without ghost instabilities, precisely because it introduces no propagating scalar degree of freedom.

The cuscuton: a non-dynamical scalar with infinite sound speed

The cuscuton was introduced by Afshordi, Chung, and Geshnizjani (hep-th/0609150, Phys. Rev. D 75, 083513, 2007) as the incompressible limit of k-essence, a scalar field theory where $c_s^2 \rightarrow \infty$ and the phase space is degenerate (zero entropy per mode). Despite the infinite propagation speed, the theory is causal because perturbations carry no microscopic information— [arXiv](https://arxiv.org/abs/hep-th/0609150) they satisfy constraint equations rather than wave equations. [ResearchGate] (https://www.researchgate.net/publication/1806701_Cuscuton_Cosmology_Dark_Energy_meets_Modified_Gravity)

The cosmological framework was developed in a companion paper (Afshordi, Chung, Doran, Geshnizjani; astro-ph/0702002, Phys. Rev. D 75, 123509, 2007), which established two crucial results. First, **cuscuton perturbations do not introduce any additional dynamical degree of freedom and only satisfy a constraint equation**, amounting to an effective modification of gravity on large scales. [arXiv +2] (https://arxiv.org/abs/astro-ph/0702002) Second, the cuscuton can realize an arbitrary equation of state—including phantom values $w < -1$ —while remaining the minimal modification of a cosmological constant. [ADS] (https://ui.adsabs.harvard.edu/abs/2007PhRvD..75I3509A/abstract) A quadratic potential reproduces Λ CDM expansion; [ADS](https://ui.adsabs.harvard.edu/abs/2007PhRvD..75I3509A/abstract) [Academia.edu] (https://www.academia.edu/79308999/Hamiltonian_analysis_of_the_cuscuton) an exponential potential reproduces DGP self-acceleration. [arXiv +2](https://arxiv.org/abs/astro-ph/0702002) Afshordi later showed (0907.5201, Phys. Rev. D 80, 081502, 2009) that the low-energy limit of non-projectable Hořava-Lifshitz gravity is uniquely the quadratic cuscuton, establishing a deep ultraviolet connection. [IAEA] (https://inis.iaea.org/records/h3k8h-9sh57)

The perturbation theory was placed on rigorous footing by Boruah, Kim, and Geshnizjani (1704.01131, JCAP 07, 022, 2017), who derived the first quadratic action for curvature perturbations ζ in cuscuton gravity. [arXiv Vanity] (https://www.arxiv-vanity.com/papers/1704.01131/) [Semantic Scholar] (https://www.semanticscholar.org/paper/Spectrum-of-Cuscuton-bounce-Kim-Geshnizjani/46a4b02592e663a29298139fa8c05bb650b56efd) The cuscuton perturbation $\delta\phi$ is eliminated algebraically via its constraint equation—substituted in terms of ζ and its first time derivative in Fourier space. This paper also provided an explicit proof of ghost-freedom around FLRW backgrounds. [arXiv Vanity +3] (https://www.arxiv-vanity.com/papers/1704.01131/) Subsequent work extended the analysis to bouncing cosmologies (Boruah, Kim, Rouben, Geshnizjani; 1802.06818, JCAP 08, 031, 2018) [ResearchGate] (https://www.researchgate.net/publication/323302724_Cuscuton_Bounce) [arXiv] (https://arxiv.org/abs/1802.06818) and, most recently, to third-order perturbations (Dehghani, Geshnizjani, Quintin; 2503.01992, JCAP 05, 026, 2025), demonstrating perturbativity and weak coupling throughout the bounce with negligible non-Gaussianities. [arXiv](https://arxiv.org/abs/2503.01992) [arXiv] (https://arxiv.org/html/2503.01992)

Three arguments for exactness of the quasi-static approximation

The quasi-static approximation in standard scalar-tensor cosmology consists of dropping time derivatives of the scalar field perturbation from the field equations, reducing a dynamical (hyperbolic) equation to an algebraic

(elliptic) constraint. In cuscuto gravity, this "approximation" is trivially exact because the scalar perturbation equation is already a constraint. Three independent lines of evidence establish this conclusion.

****The structural argument**** proceeds directly from the cuscuto Lagrangian. Since the scalar has no propagating degree of freedom, its perturbation $\delta\phi$ satisfies a constraint at each instant—there are no $\partial^2\delta\phi/\partial t^2$ or $\partial\delta\phi/\partial t$ terms to drop. [ADS](<https://ui.adsabs.harvard.edu/abs/2007PhRvD..75l3509A/abstract>) This was demonstrated explicitly by Afshordi et al. (2007) at the equation-of-motion level [OSTI +2] (<https://www.osti.gov/biblio/20935299>) and by Boruah et al. (2017) at the action level. [OSTI] (<https://www.osti.gov/biblio/22676102>) The extended cuscuto framework (Iyonaga, Takahashi, Kobayashi; 1809.10935, JCAP 12, 002, 2018) confirmed this property persists for the most general class of beyond-Horndeski theories sharing the cuscuto's two-tensor-DOF-only structure. [arXiv +2] (<https://arxiv.org/abs/1809.10935>)

****The sound-horizon argument**** leverages a general result by Sawicki and Bellini (1503.06831, Phys. Rev. D 92, 084061, 2015), who proved the QSA breaks down outside the dark energy sound horizon, [arXiv] (<https://arxiv.org/abs/1503.06831>) specifically for modes with $k < aH/c_s$. [APS Journals] (<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.92.084061>) [Unige] (<https://cosmology.unige.ch/content/limits-quasi-static-approximation-modified-gravity-cosmologies>) For theories with $c_s \rightarrow \infty$, the sound horizon encompasses all sub-Hubble scales ($k_{\text{sound}} \rightarrow 0$), and the QSA is valid everywhere. In the strict infinite-sound-speed limit of the cuscuto, the QSA is exact on all scales. [ResearchGate] (https://www.researchgate.net/publication/1806701_Cuscuto_Cosmology_Dark_Energy_meets_Modified_Gravity)

****The implementation argument**** comes from Hiramatsu and Kobayashi (2205.04688, JCAP 07, 040, 2022), who modified the CLASS Boltzmann code to constrain a cuscuto-type theory (spatially covariant gravity with only two tensor DOF and a single modified-gravity parameter β). [arXiv] (<https://arxiv.org/abs/2205.04688>) They solved the full perturbation system without invoking a QSA—because the scalar sector is already purely constraint-based. Their constraints from Planck CMB data, $-0.047 < \beta < -0.028$ at 68% CL, represent the tightest direct observational bounds on cuscuto-like gravity to date. [arXiv] (<https://arxiv.org/abs/2205.04688v1>)

The consequence is that the modified gravity parameters μ (effective gravitational coupling) and η (gravitational slip) are ****purely algebraic functions of background quantities****—scale factor, Hubble rate, and cuscuto potential parameters—with no scale dependence and no PDEs to solve in the scalar sector. Iyonaga and Kobayashi (2109.10615, Phys. Rev. D 104, 124020, 2021) confirmed that cosmological perturbation deviations from GR in these theories are parametrized entirely by time-dependent functions: [arXiv] (<https://arxiv.org/abs/2109.10615>) an effective gravitational constant $G_{\text{eff}}(t)$ and a gravitational slip. [arXiv] (<https://arxiv.org/abs/2003.01934>) A related analysis by Maeda and Panpanich (2202.04908, Phys. Rev. D 105, 104022, 2022) for cuscuto-Galileon cosmology found the perturbation equation is characterized solely by a time-dependent effective gravitational constant, with the scalar field completely determined by matter fields. [arXiv] (<https://arxiv.org/abs/2202.04908>) [APS Journals] (<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.105.104022>)

One important caveat: the cuscuto does introduce an ****elliptic (instantaneous) mode**** on each time slice (Gomes and Guariento; 1703.08226, Phys. Rev. D 95, 104049, 2017; [ResearchGate] (https://www.researchgate.net/publication/319121845_A_Class_of_Minimally_Modified_Gravity_Theories) [DOI] (<https://doi.org/10.1103/PhysRevD.97.064020>) Bhattacharyya, Coates, Colombo, Gümrukçüoğlu, Sotiriou; 1612.01824, Phys. Rev. D 97, 064020, 2018). [APS Journals] (<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.97.064020>) This is a spatial Poisson-like equation for the lapse or scalar perturbation—not a time-evolution PDE. In Fourier space, it reduces to an algebraic relation, fully consistent with the QSA being exact.

Extended cuscuto and the broader minimally modified gravity landscape

The cuscuto sits within a broader family of theories propagating only two tensor degrees of freedom—classified as "minimally modified gravity" (MMG) by Mukohyama and collaborators. [arXiv Vanity] (<https://www.arxiv-vanity.com/papers/1902.08663/>) [arXiv] (<https://arxiv.org/html/2312.06066v1>) Iyonaga, Takahashi, and Kobayashi (1809.10935, JCAP 12, 002, 2018) identified the complete class within beyond-Horndeski theory that

shares the cuscuto's non-dynamical scalar property, dubbing it **"extended cuscuto"**. [arXiv +2] (<https://arxiv.org/abs/1809.10935>) Their follow-up (2003.01934, JCAP 07, 004, 2020) studied extended cuscuto as dark energy, presenting analytic solutions mimicking Λ CDM while modifying the growth of structure through a time-dependent Newton's constant. [arXiv](<https://arxiv.org/abs/2003.01934>) [ResearchGate] (https://www.researchgate.net/publication/368400799_Consistency_of_matter_coupling_in_modified_gravity)

De Felice, Maeda, Mukohyama, and Pookkillath (2204.08294, Phys. Rev. D 106, 024028, 2022) compared two Type-IIa MMG theories—VCDM and Cuscuto—showing all cuscuto solutions are VCDM solutions (but not conversely). [arXiv +2](<https://arxiv.org/abs/2204.08294v1>) Aoki, Di Filippo, and Mukohyama (2103.15044, JCAP 05, 071, 2021) proved the cuscuto/VCDM theory genuinely differs from GR even in vacuum, with no field redefinition mapping it to Einstein gravity. [Emergent Mind](<https://www.emergentmind.com/topics/modified-massless-gravity-theory>) [arXiv](<https://arxiv.org/abs/2103.15044>)

Mylova and Afshordi (2312.06066, JHEP 04, 144, 2024) constructed an "Effective Cuscuto Theory" introducing curvature corrections systematically, confirming only two tensor DOF propagate at quadratic level. [Springer] ([https://link.springer.com/article/10.1007/JHEP04\(2024\)144](https://link.springer.com/article/10.1007/JHEP04(2024)144)) The comprehensive review by Arai, Aoki, Kobayashi et al. (2212.09094, PTEP 2023, 072E01, 2023) positions cuscuto/MMG as a distinct class within the modified gravity landscape, [arXiv](<https://arxiv.org/abs/2212.09094>) [Oxford Academic] (<https://academic.oup.com/ptep/article/2023/7/072E01/7146218>) noting the "shadowy mode" that violates the locality condition in these theories. [arXiv](<https://arxiv.org/pdf/2212.09094>)

Additional recent work includes cuscuto inflation (Bartolo, Ganz, Matarrese; 2111.06794, JCAP 05, 008, 2022), observational constraints on cuscuto inflation (Channuie, Karwan, Sangtawe; 2301.07019, Eur. Phys. J. C 83, 421, 2023), [arXiv](<https://arxiv.org/abs/2301.07019>) [Springer] (<https://link.springer.com/article/10.1140/epjc/s10052-023-11566-z>) multi-field cuscuto cosmology (Hosseini Mansoori and Molaee; 2207.06720, 2022), [arXiv](<https://arxiv.org/abs/2207.06720>) cuscuto as a limiting curvature theory (Quintin and Yoshida; 1911.06040, JCAP 02, 016, 2020), [ResearchGate] (https://www.researchgate.net/publication/337273534_Cuscuto_gravity_as_a_classically_stable_limiting_curvature_theory) [arXiv](<https://arxiv.org/abs/1911.06040>) and black hole perturbations in cuscuto-like gravity (Saito and Kobayashi; 2308.00267, Phys. Rev. D 108, 104063, 2023). [DOI Resolver] (<https://dx.doi.org/10.1103/PhysRevD.108.104063>)

Ghost-free phantom dark energy: what works and what doesn't

A rigorous no-go theorem (Vikman; astro-ph/0407107, Phys. Rev. D 72, 043527, 2005) establishes that a single canonical, minimally coupled scalar field cannot cross the phantom divide $w = -1$ without ghost instabilities. [arXiv](<https://arxiv.org/html/2508.01378>) All successful mechanisms require non-minimal couplings, higher-derivative structures, multiple fields, or modified gravity. The post-DESI era has intensified this search.

****Kinetic gravity braiding**** (Deffayet, Pujolàs, Sawicki, Vikman; 1008.0048, JCAP 10, 026, 2010) remains the pioneering ghost-free phantom crossing mechanism. [arXiv](<https://arxiv.org/abs/1008.0048>) The cubic Horndeski Lagrangian $L = K(\phi, X) + G_3(\phi, X)\Box\phi$ creates an essential mixing of scalar and tensor kinetic terms, generating effective contributions to pressure proportional to $H\dot{G}$ that allow $w < -1$ without a wrong-sign kinetic term. The late-time attractor is de Sitter. [arXiv](<https://arxiv.org/abs/1008.0048>) Linder (2512.03139, December 2025) recently showed that within shift-symmetric cubic Horndeski gravity, ****pure k-essence alone cannot cross the phantom divide****—the G_3 braiding term is essential. [arXiv](<https://arxiv.org/html/2512.03139>) Tsujikawa (2508.17231, Phys. Rev. D, 2025) proved that breaking shift symmetry (introducing $V(\phi)$ alongside Galileon self-interaction) enables phantom crossing at $z_c \approx 0.42$ without instabilities. [arXiv] (<https://arxiv.org/abs/2508.17231>) [arXiv](<https://www.arxiv.org/pdf/2508.17231>) Tiwari and Sen (2412.00931, Phys. Rev. D 111, 043530, 2025) demonstrated phantom crossing with negative dark energy densities at high redshifts in Horndeski gravity, verified with `hi_class` and MCMC analysis. [ResearchGate +2] (https://www.researchgate.net/publication/386375089_Exploring_cosmological_imprints_of_phantom_crossing_with_dynamical_dark_energy)

The ****EFT approach**** by Creminelli, D'Amico, Noreña, and Vernizzi (0811.0827, JCAP 02, 018, 2009) showed that the phantom divide can be crossed smoothly in single-field dark energy when higher-derivative (ghost-condensate-like) operators stabilize short-scale gradient instabilities. [arXiv](<https://arxiv.org/pdf/0811.0827>)

Near $w = -1$, the theory transitions between k-essence behavior ($w > -1$) and ghost-condensate behavior ($w < -1$), with **virtually zero sound speed** on cosmological scales in the phantom regime.

DHOST and **scordatura** theories offer the most general framework. Gorji, Motohashi, and Mukohyama (2009.11606, JCAP 03, 081, 2021) introduced the scordatura mechanism—controlled detuning of the DHOST degeneracy condition—to resolve the strong coupling problem of stealth dark energy solutions. [IOPscience] (<https://iopscience.iop.org/article/10.1088/1475-7516/2021/03/081>) [ADS] (<https://ui.adsabs.harvard.edu/abs/2021JCAP...03..081G/abstract>) De Felice, Mukohyama, and Takahashi (2204.02032, Phys. Rev. D, 2022) showed U-DHOST theories have scordatura built in, with the apparent Ostrogradsky mode being non-propagating ("shadowy"). [arXiv](<https://arxiv.org/abs/2204.02032>)

Hubble-dependent kinetic structures and the $K \sim 1/H^2$ question

The specific proposal of $K \propto 1/H^2$ achieving phantom expansion without ghosts has **no direct precedent** in the literature. The most relevant result is a no-go theorem by Sahoo (2601.17873, January 2026), who analyzed single-field effective scalar cosmologies with **"Hubble modulated kinetic response"** $P_\varphi = P(X, H, \varphi)$, motivated by infrared-modified and nonlocal gravitational frameworks. Imposing ghost freedom ($\partial P_\varphi / \partial X > 0$) renders the phantom divide $w = -1$ an **invariant and dynamically stable manifold** of the cosmological flow. Continuous ghost-free crossing into the phantom regime is forbidden in this class— [arXiv] (<https://arxiv.org/abs/2601.17873>) the system is attracted asymptotically to $w \rightarrow -1$ from above without crossing. [arXiv](<https://arxiv.org/html/2601.17873>) This represents a dynamical extension of earlier kinematical no-go theorems, applicable precisely to the type of Hubble-dependent kinetic structure described in the query. [arXiv](<https://arxiv.org/html/2601.17873>)

Several frameworks achieve phantom crossing through mechanisms that effectively couple kinetic structure to the Hubble rate, though not via $K \propto 1/H^2$ explicitly. The **"Freezing Gravity"** model by Yao, Ye, and Silvestri (2508.01378, August 2025) is particularly notable: within spatially covariant gravity, the equation of state is algebraically determined by two free time-functions in the Lagrangian, and the scalar's sound speed **diverges at large scales**, making the field non-dynamical in the infrared— [arXiv](<https://arxiv.org/html/2508.01378>) directly paralleling cusciton behavior. [arXiv +3](<https://arxiv.org/abs/2508.01378>) The authors explicitly note that "stable NEC violation can be achieved in Freezing Gravity, even when reduced to the Horndeski subclass, because the scalar DOF is completely suppressed and associated to the cusciton theory where the no-go theorem does not apply." [arXiv](<https://arxiv.org/pdf/2508.01378>) [arXiv](<https://arxiv.org/html/2508.01378>)

The cusciton itself can produce phantom-like dark energy. Afshordi et al. (2007) showed the cusciton can have an **arbitrary equation of state** including $w < -1$, since the cusciton field satisfies a constraint—not a dynamical equation—and introduces no ghost. [arXiv +3](<https://arxiv.org/abs/astro-ph/0702002>) Bazeia, Dantas, and Santos da Costa (2501.14909, Eur. Phys. J. C 85, 196, 2025) demonstrated that adding a cusciton-like term to a standard dark energy scalar produces a **phantom phase** ($w < -1$) that anticipates the cosmological constant phase, with strong observational support over Λ CDM via the Akaike Information Criterion. [arXiv](<https://arxiv.org/abs/2501.14909>) [Springer](<https://link.springer.com/article/10.1140/epjc/s10052-025-13855-1>)

Other relevant mechanisms include "effective phantom" from modified gravity, where $f(R)$ or Gauss-Bonnet gravity produce apparent phantom crossing without any NEC violation (Nojiri, Odintsov, Oikonomou; 2512.06279, December 2025), interacting dark energy–dark matter models where varying DM mass mimics phantom [arXiv +2](<https://arxiv.org/abs/2512.06279>) (La Penna, Redi et al.; 2601.05235, January 2026), [arXiv] (<https://arxiv.org/html/2503.16415>) and phantom dark energy's connection to time crystals in non-canonical scalar theories (Scientific Reports, 2025; 2502.08894). [Nature](<https://www.nature.com/articles/s41598-025-14048-2>)

Conclusion

The cusciton's defining property—no propagating scalar degree of freedom—has consequences that extend far beyond its original formulation. For perturbation theory, it renders the quasi-static approximation **exact** by

construction**, not through any limiting procedure but because the scalar equation is intrinsically a constraint. The modified gravity parameters μ and η become purely algebraic functions of the background, enormously simplifying observational predictions. This has been confirmed through structural analysis of the action (Boruah et al. 2017), sound-horizon arguments (Sawicki and Bellini 2015), [APS Journals] (<https://journals.aps.org/prd/abstract/10.1103/PhysRevD.92.084061>) and direct Boltzmann-code implementation (Hiramatsu and Kobayashi 2022), [arXiv](<https://arxiv.org/abs/2205.04688v1>) though the explicit statement "the QSA is exact in cuscuto gravity" appears nowhere in the literature as a standalone result—it is an inescapable synthesis of established results.

For phantom dark energy, the cuscuto and its extensions occupy a unique position: because the scalar is non-dynamical, the standard ghost no-go theorems do not apply, and arbitrary equations of state are achievable. [arXiv +3](<https://arxiv.org/abs/astro-ph/0702002>) The specific kinetic structure $K \propto 1/H^2$ has not been proposed in the literature, and Sahoo's 2026 no-go theorem suggests that simple Hubble-modulated kinetic responses in single-field Einstein gravity cannot cross the phantom divide while maintaining ghost freedom. [arXiv] (<https://arxiv.org/html/2601.17873>) [arXiv](<https://arxiv.org/abs/2601.17873>) However, the Freezing Gravity framework—where the scalar becomes cuscuto-like in the infrared—achieves phantom crossing precisely by suppressing the scalar's dynamics, [arXiv](<https://arxiv.org/pdf/2508.01378>) [arXiv] (<https://arxiv.org/html/2508.01378>) suggesting the route to ghost-free phantom expansion runs not through engineering the kinetic function's dependence on H , but through eliminating the scalar degree of freedom altogether. [arXiv](<https://arxiv.org/pdf/2508.01378>) [arXiv](<https://arxiv.org/html/2508.01378>)